Verification 09_HM_Simulator_FEM_FVM_2D

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1 Objective

The purpose of this verification exercise is to demonstrate that the numerical simulator correctly reproduces the solution of Mandel's problem.

2 Governing Equations

2.1 Fluid flow

The fluid-flow model solves the slightly compressible single-phase flow equation,

$$\frac{\partial(\phi\rho)}{\partial t} = \nabla \cdot \left(\frac{\rho k}{\mu}\nabla P\right) + \frac{Q}{V}, \tag{1}$$

where P is pore pressure, ρ is fluid density, ϕ is porosity, k is permeability, μ is viscosity, V is the cell volume, and Q represents sources and sinks.

2.2 Mechanics

The mechanical equilibrium equation is

$$\nabla \cdot \boldsymbol{\sigma} + \mathbf{b} = \rho\ddot{\mathbf{u}}, \tag{2}$$

where $\boldsymbol{\sigma}$ is the Cauchy stress tensor, $\mathbf{b}$ is the body force per unit volume, ρ is the material density, and $\mathbf{u}$ is the displacement vector. For the quasi-static problems considered here, inertial effects are neglected, giving

$$\nabla \cdot \boldsymbol{\sigma} + \mathbf{b} = \mathbf{0}. \tag{3}$$

For an isotropic linear elastic material, the stress–strain relationship is

$$\begin{bmatrix}\sigma_{xx}\\ \sigma_{yy}\\ \tau_{xy}\end{bmatrix} = \mathbf{D}\begin{bmatrix}\varepsilon_{xx}\\ \varepsilon_{yy}\\ \gamma_{xy}\end{bmatrix}, \tag{4}$$

where the constitutive matrix $\mathbf{D}$ depends on whether a plane stress or plane strain assumption is adopted. For plane stress,

$$\mathbf{D} = \frac{E}{1-\nu^2}\begin{bmatrix}1 & \nu & 0\\ \nu & 1 & 0\\ 0 & 0 & \dfrac{1-\nu}{2}\end{bmatrix}, \tag{5}$$

whereas for plane strain,

$$\mathbf{D} = \frac{E}{(1+\nu)(1-2\nu)}\begin{bmatrix}1-\nu & \nu & 0\\ \nu & 1-\nu & 0\\ 0 & 0 & \dfrac{1-2\nu}{2}\end{bmatrix}. \tag{6}$$

3 Mandel's Solution

Mandel's problem (Mandel, 1953), Figure 1, consists of a poroelastic block which is homogeneous and isotropic. It is wedged between two plates which suddenly apply a force, F, with units of Newton per distance in the z-direction. This force results in a sudden increase in pressure, termed the Mandel-Cryer effect. The response of the block to this force has been described by Castelletto et al. (2015), whose approach will be followed here.

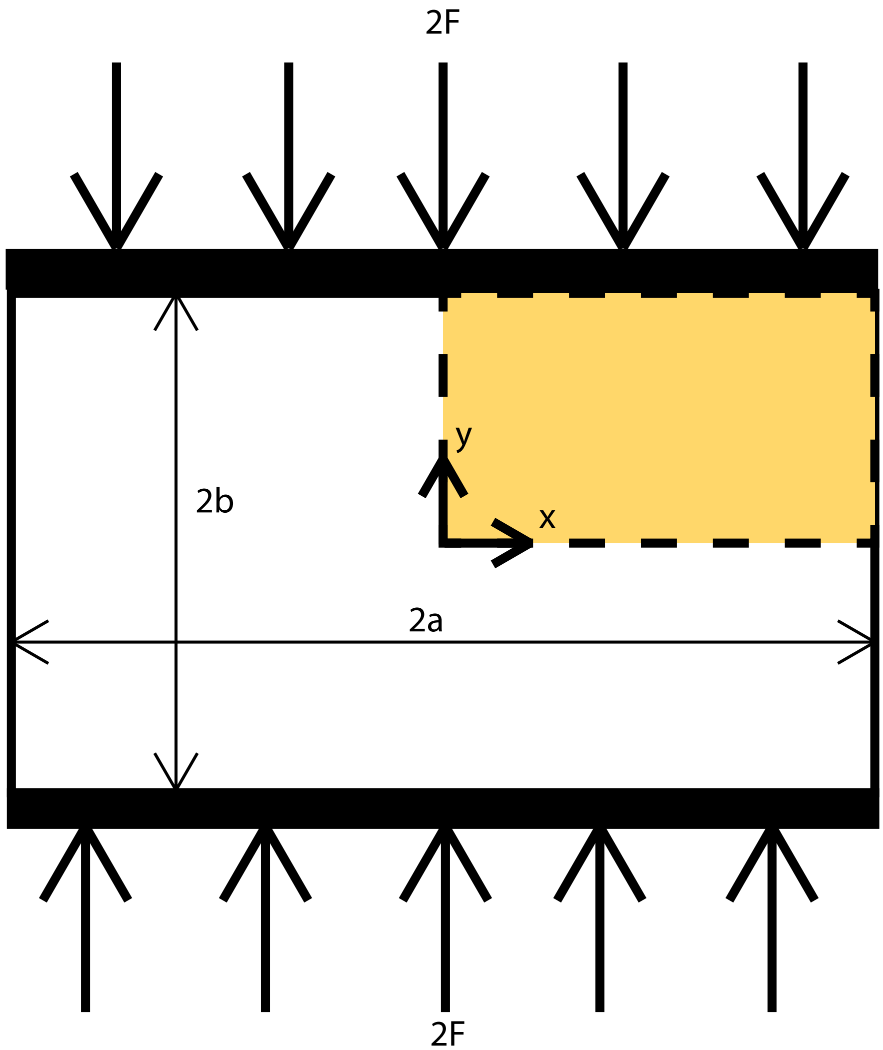

Figure 1: An overview of Mandel's problem.

Due to symmetry, only a quarter of the $2a$ by $2b$ block needs to be modelled. First, certain key parameters are calculated such as the shear modulus, $G = \frac{E}{2(1-2\nu)}$, noting that the Poisson's ratio is the drained variant, the uniaxial drained bulk modulus, $K_v = \frac{2G(1-\nu)}{1-2\nu}$, the bulk modulus, $K = \frac{2G(\nu+1)}{3(1-2\nu)}$, a mobility, $k_m = \frac{k}{\mu}$, and the Biot coefficient, $\alpha = 1 - \frac{K}{K_s}$. Next, the fluid bulk modulus, $K_f = c_f^{-1}$, Biot's modulus, $M = (\frac{\alpha-\phi}{K_s} + \frac{\phi}{K_f})^{-1}$, and the Skempton coefficient, $B = \frac{\alpha M}{K+\alpha^2 M}$, are also found. Following this, the undrained Poisson's ratio, $\nu_u = \frac{B\alpha(1-2\nu)+3\nu}{3-B\alpha(1-2\nu)}$, the fluid diffusivity coefficient, $c_v = \frac{k_m M K_v}{K_v+\alpha^2 M}$, and an auxillary elastic coefficient, $\xi = \frac{1-\nu}{\nu_u-\nu}$, are found.

These parameters are used to find all of the positive roots, β_n, that satisfy (Castelletto et al., 2015),

$$\tan\left(\beta_n\right) = \xi \beta_n \tag{7}$$

Dimensionless pore pressure can the be found using (Castelletto et al., 2015)

$$\frac{P}{P_0} = 2 \sum_{n=1}^{\infty} \frac{\sin\left(\beta_n\right)}{\beta_n - \sin\left(\beta_n\right)\cos\left(\beta_n\right)} \left(\cos\left(\frac{\beta_n x}{a}\right)\cos\left(\beta_n\right)\right)\exp\left(-\frac{\beta_n^2 c_v t}{a^2}\right), \tag{8}$$

with (Castelletto et al., 2015)

$$P_0 = \frac{B}{3a}\left(1+\nu_u\right)F. \tag{9}$$

These solutions match well with the simulators, Figures 2, **??**, and xxx.

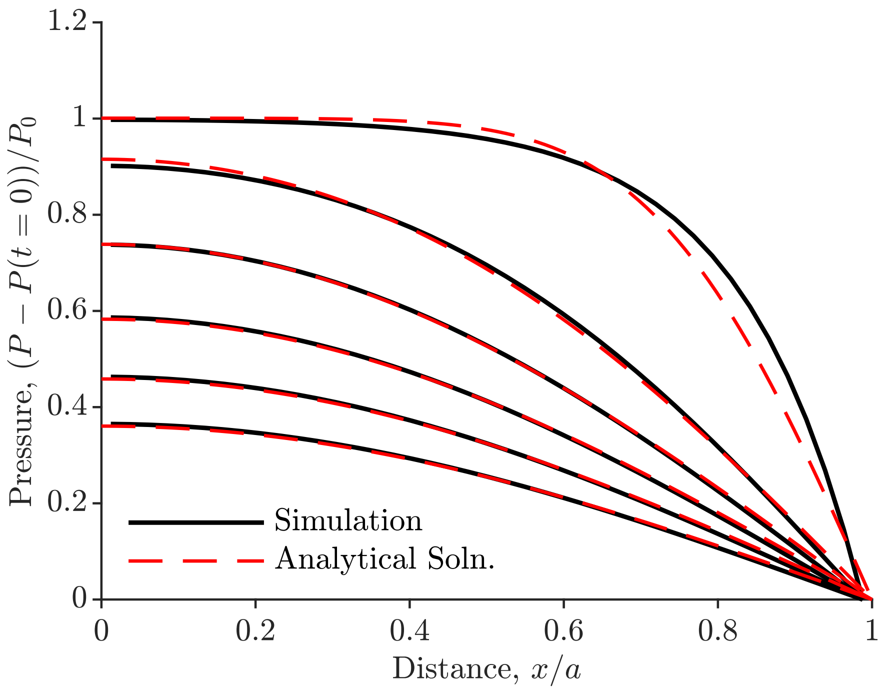

Figure 2: The MATLAB simulator results shown against the Mandel solution for various times. Note that the simulator artificially sets the right-hand boundary to zero pressure using wells in the right-most cell, potentially resulting in slight numerical error. The solutions for the simulator are plotted for a height of 3a/2.

The displacement in the x-direction can be calculated using (Castelletto et al., 2015),

$$u_x = \left[\frac{F\nu}{2Ga} - \frac{F\nu_u}{Ga}\sum_{n=1}^{\infty}\frac{\sin(\beta_n)\cos(\beta_n)}{\beta_n - \sin(\beta_n)\cos(\beta_n)} \times \exp\left(-\frac{\beta_n^2 c_f t}{a^2}\right)\right] x + \frac{F}{G}\sum_{n=1}^{\infty}\frac{\cos(\beta_n)}{\beta_n - \sin(\beta_n)\cos(\beta_n)}\sin\left(\frac{\beta_n x}{a}\right) \times \exp\left(-\frac{\beta_n^2 c_f t}{a^2}\right). \tag{10}$$

This result matches satisfactorily with both simulators, Figures 4 and xxx.

Finally, the displacement in the y-direction can be calculated using (Castelletto et al., 2015),

$$u_y = \left[-\frac{F\left(1-\nu\right)}{2Ga} + \frac{F\left(1-\nu_u\right)}{Ga}\sum_{n=1}^{\infty}\frac{\sin\left(\beta_n\right)\cos\left(\beta_n\right)}{\beta_n - \sin\left(\beta_n\right)\cos\left(\beta_n\right)}\exp\left(-\frac{\beta_n^2 c_v t}{a^2}\right)\right] y. \tag{11}$$

This result matches well with both simulators, Figures 5 and xxx.

The simulation parameters used for the test are given in Table 1.

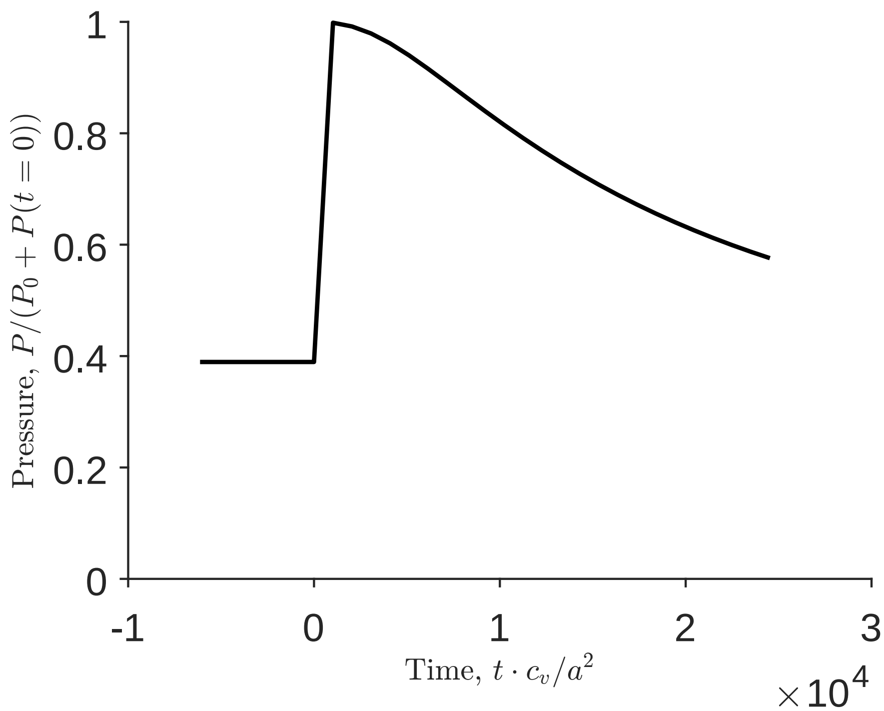

Figure 3: The MATLAB simulator results. Note the Mandel-Cryer effect clearly visible at $t = 0$. Additionally the jump in pressure matches the P_0 predicted for this problem.

4 Discussion

The numerical results reproduce Mandel's solution for the compressible case. This test verifies the implementation of the constitutive law, equilibrium equations, boundary conditions, and coupling.

5 Conclusion

These benchmark problems verify the implementation of the poroelastic simulator.

References

N. Castelletto, J. White, and H. Tchelepi. Accuracy and convergence properties of the fixed-stress iterative solution of two-way coupled poromechanics. *International Journal for Numerical and Analytical Methods in Geomechanics*, 39:1593–1618, 2015. https://dx.doi.org/10.1002/nag.2400.

J. Mandel. Consolidation des sols (etude mathématique). *Géotechnique*, 3:287–299, 1953. https://doi.org/10.1680/geot.1953.3.7.287.

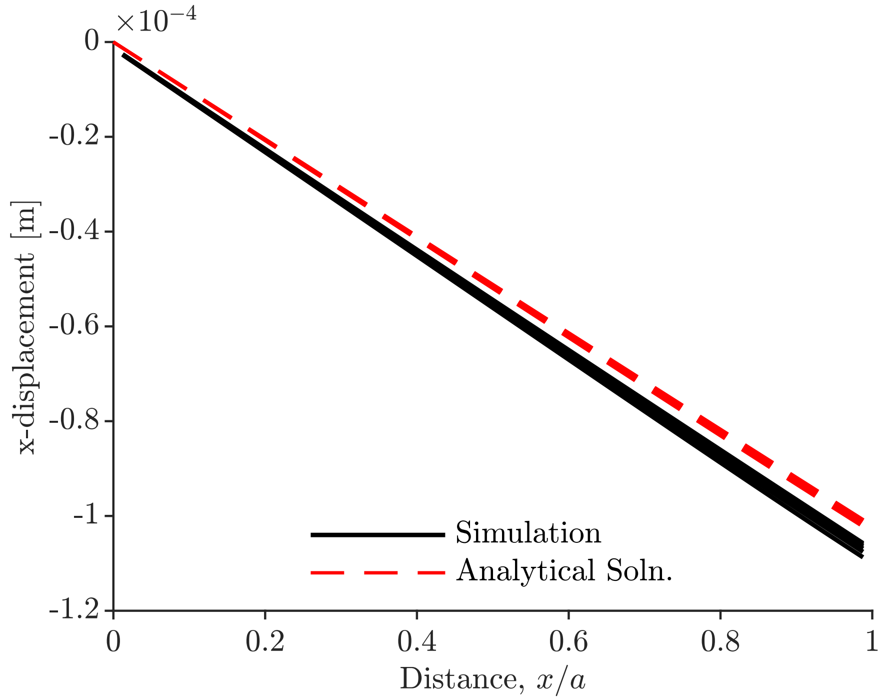

Figure 4: The MATLAB simulator results shown against the Mandel solution for various times.

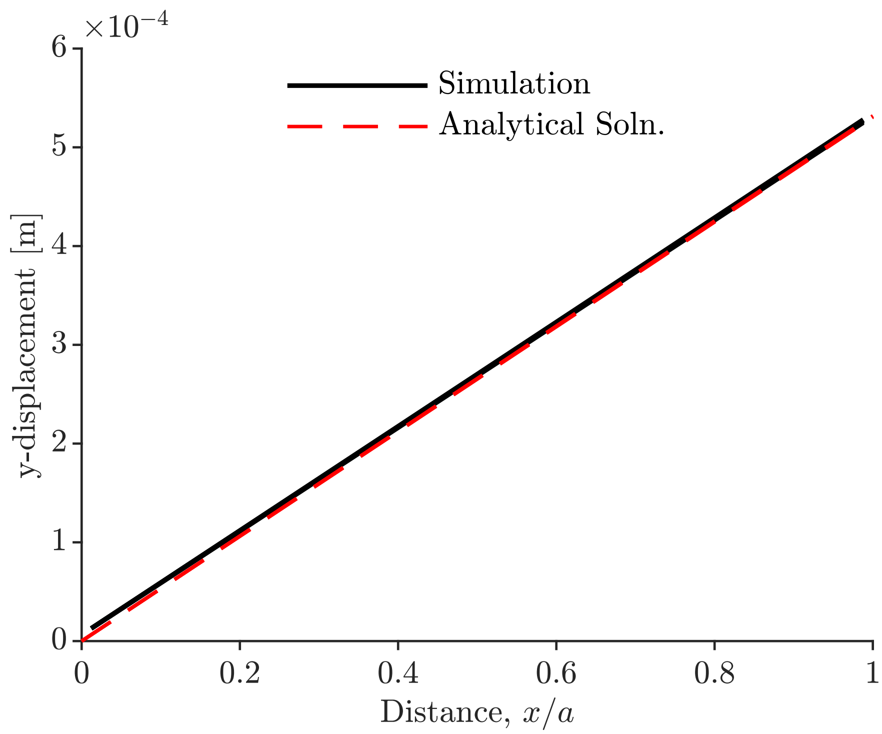

Figure 5: The MATLAB simulator results shown against the Mandel solution for various times.

Parameter	Symbol	Value
Final time	t	200 s
Number of elements	N_x	41
Number of elements	N_y	41
Domain length	L_x	1 m
Domain width	L_y	1 m
Young's Modulus	E	18.3×10^9 Pa
Poisson's ratio	ν	0.16
Grain Bulk Modulus	K_s	36×10^9 Pa
Permeability	k	1×10^{-12} m^2
Initial porosity	ϕ	0.2
Fluid compressibility	c_f	1×10^{-8} Pa^{-1}
Fluid viscosity	μ_f	0.1 Pa ·s
Initial fluid density	ρ_0	1000 kg/m^3
Load on top edge	F	10×10^6 N/m

Table 1: Simulation parameters for test